

Partitions

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September 2026

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1 Introduction to Partitions

1.1 Basic Definitions

(1.1.1) **Definition:** A partition of $n \in \mathbb{N}$ is a way of writing n as the sum of numbers. These will be denoted in this document as $(p_1 + p_2 + \dots + p_i) \vdash n$.

(1.1.2) **Definition:** A part is one of the numbers that make up a partition. It has an associated part size, which is the value of the number.

(1.1.3) **Definition:** The partition function of n is the number of possible partitions of n . This will be denoted as $p(n) = i$.

(1.1.4) **Definition:** The set of partitions is the set containing all possible partitions of n . This will be denoted using non-standard notation as $\overline{P}(n) = \{p_1, p_2, \dots, p_i\}$ where p_x is a partition of n and $i = p(n)$.

(1.1.5) **Definition:** Conditional partitions can also be created which limit which partitions are valid based on a given condition. This can be applied to both partition functions and sets of partitions as $p(n|\text{condition})$ and $\overline{P}(n|\text{condition})$.

1.2 Bijective Proofs

(1.2.1) **Definition:** A bijection is a mathematical function that is a one-to-one and onto mapping (Merriam-Webster, 2025).

We can prove that there is a bijection between two sets A and B by defining the functions $f : A \rightarrow B$ and $f^{-1} : B \rightarrow A$. This can be useful when trying to compare the size of two partition functions that would be hard to compute as if there is a bijection between two sets of partitions, the corresponding partition functions will be equal.

(1.2.2) **Example:** For example, we can prove that, for any $n \geq 1$, $n \in \mathbb{N}$,

$$p(n|\text{all parts are even}) = p(n|\text{even number of parts}) \quad (1.2.3)$$

Proof: To prove Equation 1.2.3, we let $A = P(n|\text{all parts are even})$ and $B = P(n|\text{even number of parts})$ such that $|A|$ and $|B|$ equal the sets' respective partition functions. We then define the functions $f : A \rightarrow B$ and $f^{-1} : B \rightarrow A$ as:

- $f : A \rightarrow B$: For each partition $a \in A$, take each of its constituent parts, a_i , and replace each of them with two parts half the size $(\frac{1}{2}a_i + \frac{1}{2}a_i)$. The resulting partition should be a member of B . i.e.,

$$f : (2a_1 + 2a_2 + \dots + 2a_n) \rightarrow (a_1 + a_1 + a_2 + a_2 + \dots + a_n + a_n)$$

- $f^{-1} : B \rightarrow A$: For each partition $b \in B$, take each of its constituent parts, b_i , and pair each of them with one of equal size. Each pair of parts $(b_i + b_i)$ should then be replaced with a single part, $2b_i$, double the size of either. The resulting partition should be a member of A . i.e.,

$$f^{-1} : ((a_1 + a_1) + (a_2 + a_2) + \dots + (a_n + a_n)) \rightarrow (2a_1 + 2a_2 + \dots + 2a_n)$$

Then, to prove $|A| = |B|$, we take a partition from A and apply f and f^{-1} , the resulting partition will be the original partition we started with. This means there must be a bijection between A and B , and therefore Equation 1.2.3 holds for any $n \geq 1$, $n \in \mathbb{N}$.

□

1.3 Euler's Identity

(1.3.1) **Theorem:** Euler's Identity states that, for any positive integer n ,

$$p(n|\text{each part is odd}) = p(n|\text{each part is distinct}) \quad (1.3.2)$$

This identity can be further generalised to get

(1.3.3) **Theorem:** For any pair of integers, $k \geq 2$, $n \geq 1$,

$$p(n|\text{no part is a multiple of } k) = p(n|\text{each part is repeated at most } (k-1) \text{ times}) \quad (1.3.4)$$

Proof: Let $A = P(n|\text{no part is a multiple of } k)$ and $B = P(n|\text{each part is repeated at most } k-1 \text{ times})$. We then define $f : A \rightarrow B$ and $f^{-1} : B \rightarrow A$ as:

- $f : A \rightarrow B$: Take a partition from A . Replace any groups of equally sized parts which appear at least k times with a new part with size k -times larger than the original size. Once no part size appears more than $k-1$ times, the resulting partition will be a member of B . i.e.,

$$f : ((x_1 + x_2 + \dots + x_k) + y + \dots) \rightarrow (kx + y + \dots)$$

- $f^{-1} : B \rightarrow A$: Take a partition from B . Any parts with size kx , $x \in \mathbb{N}^{>0}$ are replaced with k -length tuples of parts who are k -times smaller. After all part size multiples of k are replaced, the resulting partition will be a member of A . i.e.,

$$f^{-1} : (kx + y + \dots) \rightarrow ((x_1 + x_2 + \dots + x_k) + y + \dots)$$

Then, to prove $|A| = |B|$, we take a partition from A and apply f and f^{-1} , the resulting partition will be the original partition we started with proving that there must be a bijection between A and B , and therefore Equation 1.3.4 holds for any pair of integers, $k \geq 2$, $n \geq 1$. Euler's Identity is a specific example of Theorem 1.3.3 and therefore, must also be true. □

1.4 Euler Pairs and Alder's Conjecture

We recall the following definition and theorem from (Subbarao, 1971),

(1.4.1) **Definition:** An Euler pair of order r is a pair of sets of integers S_1 and S_2 satisfying, for all $n \geq 1$, $n \in \mathbb{N}$,

$$p(n|\text{all parts are in } S_2) = p(n|\text{all parts are repeated at most } (r-1) \text{ times and in } S_1) \quad (1.4.2)$$

(1.4.3) **Theorem:** (S_1, S_2) is an Euler pair of order r if and only if

$$rS_1 \subseteq S_1 \text{ and } S_2 = S_1 - rS_1 \quad (1.4.4)$$

e.g., $S_2 = \{s \in S_1 | s \notin rS_1\}$.

Proof: Theorem 1.4.3 is proved comprehensively in (Subbarao, 1971, pp. 333-335). □

(1.4.5) **Conjecture:** From (Durham University, 2026), Alder's Conjecture is defined as

$$p(n|\text{all parts are in } S_2) = p(n|\text{all } k\text{-distinct parts}) \quad (1.4.6)$$

which has been proven false for all $k \geq 3$. However, (Yee, 2008, p. 1) instead states that the number of partitions of n into parts differing by d is greater than or equal to that of partitions of n into parts $\equiv (\text{mod } d+3)$. This definition of the conjecture was still declared open when this paper was published, and has a greater number of valid cases.

2 Visualising Partitions

2.1 Young Diagrams and Ferrers Graphs

(2.1.1) **Definition:** Young diagrams represent partitions by stacking rows of squares (corresponding to individual parts) on top of each other, in ascending order (See Figure 1).

(2.1.2) **Definition:** Ferrers graphs represent partitions in a similar way, but instead electing to utilise dots instead of squares when representing parts.

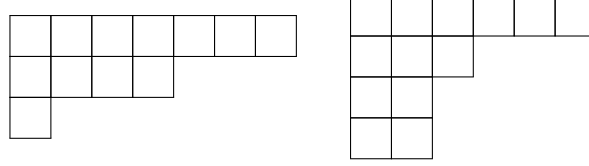


Figure 1: Young diagrams ($n = 13$)

From this point, I will be using Young diagrams as my preferred choice of visualisation, but Ferrers graphs would be equally valid.

2.2 Solving Problems with Visualisations

Example 2.2.1: Previously, in Example 1.2.2, we used a bijection to prove that, for any $n \geq 1, n \in \mathbb{N}$,

$$p(n | \text{all parts are even}) = p(n | \text{even number of parts}) \quad (1.2.3 \text{ (Repeated)})$$

Proof: This can similarly be achieved using Ferrers graphs by redefining f and f^{-1} to be the following:

- $f : A \rightarrow B$: Take a partition, $a \in A$, split each constituent part (row of the graph) down the middle, and form two new equally sized rows. This resulting partition will be a member of B .
- $f^{-1} : B \rightarrow A$: Take a partition, $b \in B$, and combine pairs of equally sized parts into single parts double the size of the originals. This results in a partition from A .

As we can see in Figure 2, when the functions are applied in turn to an example partition from A , the partition becomes a member of B , before being transformed back to the original partition. This demonstrates the bijection between the sets and again proves Equation 1.2.3.

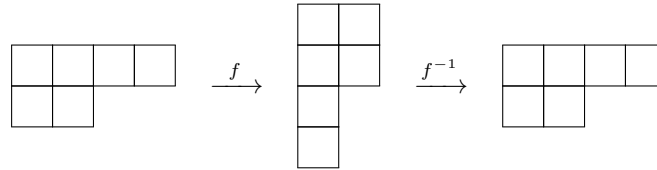


Figure 2: Applying f and f^{-1}

□

2.3 Partition Conjugates

(2.3.1) **Definition:** A conjugate of a partition is a partition which has had its rows and columns transposed to give a new partition.

(2.3.2) **Definition:** A self-conjugate partition is a partition which is a conjugate of itself; meaning that the original partition is the same as the transposed version.

Example 2.3.3: This can produce some interesting results, such as proving that, for any $n \geq 1, n \in \mathbb{N}$,

$$p(n|\text{self-conjugate}) = p(n|\text{all parts are both distinct and odd}) \quad (2.3.4)$$

Proof: Let $A = P(n|\text{self-conjugate})$ and $B = P(n|\text{all parts are both distinct and odd})$. We then define $f : A \rightarrow B$ and $f^{-1} : B \rightarrow A$ as:

- $f : A \rightarrow B$: Take a partition from A . Recursively strip away the upper-left most row and column of the graph and form a new part of size $2s - 1$ where s is the length of both the row and column. Once this process is finished, the resulting partition will be a member of B .
- $f^{-1} : B \rightarrow A$: Take a partition from B . Take each constituent part of length s and use it to form a self-conjugate pattern with height and width $\frac{s}{2} + 1$. Nest this pattern inside the previous one (or on its own for the first). Once this has been done to all the rows, the resulting partition will be a member of A .

As we can see in Figure 3, when the functions are applied in turn to an example partition from A , the partition becomes a member of B , before being transformed back to the original partition. This demonstrates the bijection between the sets and again proves Equation 2.3.4.

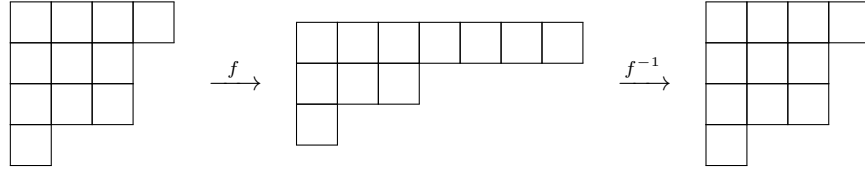


Figure 3: Applying f and f^{-1}

□

2.4 Durfee Squares

(2.4.1) **Definition:** The Durfee square of a Young diagram is the largest square that fits inside the diagram. It has size s if it has sides of length s .

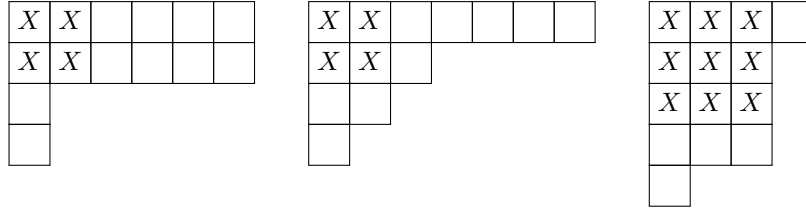


Figure 4: Durfee squares

(2.4.2) **Theorem:** The number of partitions of n having a Durfee square of size s can be written as

$$\sum_{m=0}^{n-s^2} p(m|\text{all parts are } \leq s) p(n-m-s^2|\text{all parts are } \leq s) \quad (2.4.3)$$

Proof: Let $A = P(n|\text{Durfee square of size } s)$. We can infer from 2.4.1 that the Durfee square will contribute s^2 towards the total area of a partition from A , meaning the remaining area is $n - s^2$. This remaining area can then be split into the area to the right of the square, A_r , and the area underneath, A_u .

Let the area of A_u be $0 \leq m \leq n - s^2$. We know that, for the Durfee square to be maximal, the constituent parts of A_u must be at most s , so we get that the number of partitions of A_u for each size of Durfee square is $p(m|\text{all parts are } \leq s)$, meaning the total number of possible partitions is

$$p(A_u) = \sum_{m=0}^{n-s^2} p(m|\text{all parts are } \leq s) \quad (2.4.4)$$

This leaves the area of A_r to be $n - m - s^2$. As the Durfee square is maximal, the $(s + 1)^{th}$ column must also be $\leq s$, and as the parts are ordered, this also restricts the parts of A_r to be $\leq s$. As a result, the number of partitions for each size of Durfee square is $p(n - m - s^2 | \text{all parts are } \leq s)$, and when combined with Equation 2.4.4, we get Equation 2.4.3, proving Theorem 2.4.2.

□

3 Generating Functions

3.1 Creating Generating Functions for Partitions

(3.1.1) **Theorem:** The values of $p(n)$ can be calculated, for any $n \geq 0, n \in \mathbb{N}$, by computing the coefficients of the generating function

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{j=1}^{\infty} \frac{1}{1-x^j} \quad (3.1.2)$$

Proof: Given that, for some value of $a \geq 1, n \geq 0, a, n \in \mathbb{N}$,

$$\frac{1}{1-x^a} = (x^0 + x^a + x^{2a} + x^{3a} + \dots + x^{ia} + \dots) = \sum_{n=0}^{\infty} x^{na} \quad (3.1.3)$$

the powers of x generated are the values which can be represented as partitions solely made from parts of size a . e.g., $a = 3, \frac{1}{1-x^3} = 1 + x^3 + x^6 + x^9 + \dots x^{3n} + \dots$

Let $p_k(n)$ be the number of partitions of n using parts of at most size k . Based on Equation 3.1.3, and that $(x^\alpha)(x^\beta) = x^{\alpha+\beta}$, we can formulate that, for example,

$$\sum_{n=0}^{\infty} p_2(n)x^n = \prod_{j=1}^2 \frac{1}{1-x^j} = (1 + x + x^2 + \dots)(1 + x^2 + x^4 + \dots) = 1 + x + 2x^2 + \dots \quad (3.1.4)$$

which can be generalised to be

$$\sum_{n=0}^{\infty} p_k(n)x^n = \prod_{j=1}^k \frac{1}{1-x^j} \quad (3.1.5)$$

As $k \rightarrow \infty, p_k(n) \rightarrow p_\infty(n)$. With this limit, partitions of n are created using parts of unbounded size, which is equivalent to $p(n)$, establishing Theorem 3.1.1.

□

3.2 Solving Problems with Generating Functions

(3.2.1) **Theorem:** Given a pair of values $(n, a) \in \mathbb{N}^2, n \geq a$, the partition function $p(n | \geq 1 \text{ part of size } a)$ can be computed using

$$\sum_{n=0}^{\infty} p(n | \geq 1 \text{ part of size } a)x^n = \left(\prod_{i=1}^{a-1} \frac{1}{1-x^i} \right) \left(\frac{x}{1-x^a} \right) \left(\prod_{j=a+1}^{\infty} \frac{1}{1-x^j} \right) \quad (3.2.2)$$

Proof: Equation 3.1.3 can be adjusted to require at least one part of size a ,

$$\sum_{n=1}^{\infty} x^n = x^a + x^{2a} + \dots = \frac{x}{1-x^a} \quad (3.2.3)$$

This removes the x^0 term, hence restricting the partition to have at least one part of size a . This adjustment is then incorporated into Equation 3.1.2 to produce Equation 3.2.2 and therefore proves Theorem 3.2.1. $\square$

(3.2.4) **Theorem:** Given a pair of values $(n, k) \in \mathbb{N}^2$, $n \geq k$, the partition function $p(n| \text{exactly } k \text{ parts})$ can be computed using

$$\sum_{n=0}^{\infty} p(n| \text{exactly } k \text{ parts}) x^n = \frac{x^k}{(1-x)(1-x^2)(1-x^3)\dots(1-x^k)} \quad (3.2.5)$$

Proof: Let A be a partition of n with exactly k parts and A^T be the transposed Ferrers graph of the partition. A^T will be guaranteed to have at least one part of size k , which will also be the largest part. We can therefore equate Equation 3.2.5 to $p_k(n| \geq 1 \text{ part of size } k)$. Combining Equation 3.1.5 and Theorem 3.2.1, we get that

$$\sum_{n=0}^{\infty} p(n| \text{exactly } k \text{ parts}) x^n = \left(\prod_{i=1}^{a-1} \frac{1}{1-x^i} \right) \left(\frac{x}{1-x^a} \right) \left(\prod_{j=a+1}^k \frac{1}{1-x^j} \right) \quad (3.2.6)$$

which is equivalent to Equation 3.2.5, proving Theorem 3.2.4. $\square$

3.3 Revisiting Euler's Identity

Earlier, we defined Euler's Identity in Theorem 1.3.1. We can now easily prove this using generating functions.

$$p(n| \text{each part is odd}) = p(n| \text{the parts are distinct}) \quad (3.3.2)$$

Proof: We can adapt the generating function for partitions given by Theorem 3.1.1 to create a generating function for $p(n| \text{each part is odd})$ by excluding the even powers of x . We accomplish this by dividing the unchanged generating function by the generating function for partitions made from even parts,

$$\left(\prod_{j=1}^{\infty} \frac{1}{1-x^j} \right) \left(\prod_{j=1}^{\infty} \frac{1}{1-x^{2j}} \right)^{-1} = \prod_{j=1}^{\infty} (1+x^j) \quad (3.3.3)$$

Similarly, for $p(n| \text{the parts are distinct})$ we can define the term expansion as

$$(1+x)(1+x^2)(1+x^3)\dots$$

as each term can either appear 0 or 1 times. When expanded for all $n \geq 1, n \in \mathbb{N}$, we get

$$(1+x)(1+x^2)(1+x^3)\dots(1+x^n)\dots = \prod_{j=1}^{\infty} (1+x^j) \quad (3.3.4)$$

As the generating functions for these two partitions are equal, their corresponding partition functions must also be equal, proving Theorem 1.3.1. $\square$

As seen earlier in Theorem 1.3.3, we can generalise Equation 1.3.2 into Equation 1.3.4 to work with multiples of k instead of odd numbers. This can again be proved using generating functions by using a similar method to the above proof.

$$p(n|no\ part\ is\ a\ multiple\ of\ k) = p(n|each\ part\ is\ repeated\ at\ most\ (k-1)\ times) \quad (1.3.4\ (Repeated))$$

Proof: We can adapt Equation 3.3.3 to exclude multiples of k rather than even numbers (multiples of 2), by adjusting the second generating function.

$$\left(\prod_{j=1}^{\infty} \frac{1}{1-x^j} \right) \left(\prod_{j=1}^{\infty} \frac{1}{1-x^{kj}} \right)^{-1} = \prod_{j=1}^{\infty} \frac{1-x^{kj}}{1-x^j} \quad (3.3.5)$$

Similarly, we can adapt Equation 3.3.4 to allow for $(k-1)$ repeats of each term. To do this, we define the term expansion as

$$(1+x+x^2+\dots+x^{(k-1)}) (1+x^2+x^4+\dots+x^{2(k-1)}) (1+x^3+x^6+\dots+x^{3(k-1)}) \dots$$

which can be rewritten as

$$\left(\frac{1+x+x^2+\dots+x^k}{1-x} \right) \left(\frac{1+x^2+x^4+\dots+x^{2k}}{1-x^2} \right) \left(\frac{1+x^3+x^6+\dots+x^{3k}}{1-x^3} \right) \dots$$

When expanded for all $n \geq 1, n \in \mathbb{N}$, we get

$$\left(\frac{1+x+x^2+\dots+x^k}{1-x} \right) \left(\frac{1+x^2+x^4+\dots+x^{2k}}{1-x^2} \right) \dots = \prod_{j=1}^{\infty} \left(\frac{1-x^{kj}}{1-x^j} \right) \quad (3.3.6)$$

As Equation 3.3.5 $\equiv$ Equation 3.3.6, we can prove Theorem 1.3.3.

□

4 Computing $p(n)$

4.1 Euler's Discovery and Franklin's Construction

(4.1.1) **Definition:** The slope, s , of a partition is the length of the longest 45° diagonal on a Ferrers diagram as seen in Figure 5.

(4.1.2) **Definition:** The base, b , of a partition is the size of the smallest part on a Ferrers diagram as seen in Figure 5.

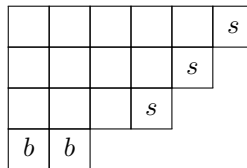


Figure 5: Partition slope and base

(4.1.3) **Definition:** Franklin's Construction is defined as one of two cases,

- If $s > b$, then the base of the partition should be removed and used to form a new slope, as seen in Figure 6.

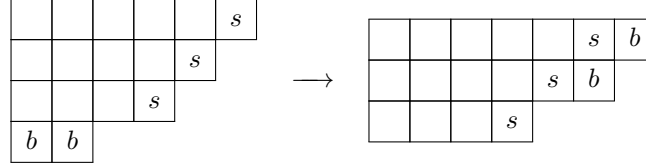


Figure 6: Case 1

- If $s < (b - 1)$, then the slope of the partition should be removed and used to form a new base, as seen in Figure 7.

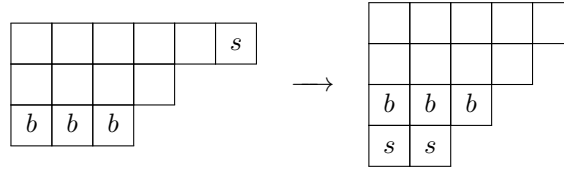


Figure 7: Case 2

(4.1.4) **Conjecture:** Euler conjectured that, for most $n \geq 0$, $n \in \mathbb{N}$,

$$p(n|\text{odd number of distinct parts}) = p(n|\text{even number of distinct parts}) \quad (4.1.5)$$

Proof: Let $p_o(n) = p(n|\text{odd number of distinct parts})$ and $p_e(n) = p(n|\text{even number of distinct parts})$. This conjecture can be proven valid for all $s \neq (b - x)$, $x = \{0, 1\}$ using Franklin's Construction as when it is applied, the parity of a partition is flipped, providing a mapping between $p_o(n)$ and $p_e(n)$. The construction breaks down one of two cases,

- When the slope is equal to the base, they will share a point, meaning the mapping is invalid as it would leave missing cells in the diagram (See Figure 8).

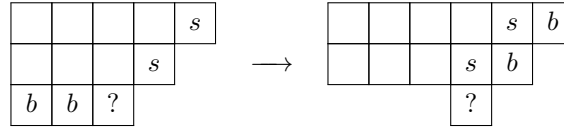


Figure 8: Invalid Case 1

- When the slope is exactly one fewer than the base, moving the slope will create an invalid Young diagram as seen in Figure 9 as the parts must be distinct.

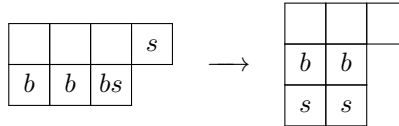


Figure 9: Invalid Case 2

These exceptional cases occur when n is a generalised pentagonal number and the slope, s , satisfies

$$n = \frac{s(3s \pm 1)}{2} \quad (4.1.6)$$

□

With this in mind, Conjecture 4.1.4 can be refined to produce,

4.2 Euler's Pentagonal Number Theorem

(4.2.2) **Theorem:** Euler's Pentagonal Number Theorem defines the piecewise function, for all $n \geq 0$, $n \in \mathbb{N}$,

$$\epsilon(n) = p_e(n) - p_o(n) = \begin{cases} (-1)^k & n = \frac{k(3k \pm 1)}{2}, k \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases} \quad (4.2.3)$$

Proof: From Equation 3.3.4 we have the generating function for distinct parts. We can adapt the equation to account for parity by representing each partition of n as $(-1)^{no. parts} x^n$. This produces the generating function

$$\sum_{n=0}^{\infty} (p_e(n) - p_o(n)) x^n = \prod_{j=1}^{\infty} (1 - x^j) = (1 - x)(1 - x^2) \dots \quad (4.2.4)$$

Euler discovered that this equation could be combined with Equation 4.1.6 to get

$$\prod_{j=1}^{\infty} (1 - x^j) = 1 + \sum_{k=1}^{\infty} (-1)^k x^{\frac{1}{2}k(3k+1)} + \sum_{k=1}^{\infty} (-1)^k x^{\frac{1}{2}k(3k-1)} \quad (4.2.5)$$

and by combining the two summations, we get

$$\sum_{n=0}^{\infty} \epsilon(n) x^n = \prod_{j=1}^{\infty} (1 - x^j) = \sum_{k=-\infty}^{\infty} (-1)^k x^{\frac{1}{2}k(3k-1)} \quad (4.2.6)$$

This generating function produces values for $\epsilon(n)$ that match with the piecewise function in Equation 4.2.3, proving Theorem 4.2.2. □

4.3 A Recurrence Relation for $p(n)$ and Ramanujan's Congruences

(4.3.1) **Theorem:** Euler wanted to find a more efficient way to compute a value for $p(n)$. Taking into account the Pentagonal Number Theorem, Euler theorised that,

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \dots \quad (4.3.2)$$

Proof: Taking Equation 3.1.2, we can multiply both sides by $\prod_{j=1}^{\infty} (1 - x^j)$ to produce

$$\left(\sum_{n=0}^{\infty} p(n) x^n \right) \left(\prod_{j=1}^{\infty} (1 - x^j) \right) = 1 \quad (4.3.3)$$

Then, using the Pentagonal Number Theorem, we can produce

$$\left(\sum_{n=0}^{\infty} p(n) x^n \right) \left(\sum_{k=-\infty}^{\infty} \left((-1)^k x^{\frac{k(3k-1)}{2}} \right) \right) = 1 \quad (4.3.4)$$

which can be multiplied out to get

$$\sum_{n=0}^{\infty} \left(\sum_{k=-\infty}^{\infty} (-1)^k p \left(n - \frac{k(3k-1)}{2} \right) \right) x^n = 1 \quad (4.3.5)$$

then, equating coefficients of x^n , for $n = 0$ gives $p(0) = 1$, and for $n > 0$, gives

$$\sum_{k=-\infty}^{\infty} (-1)^k p \left(n - \frac{k(3k-1)}{2} \right) = 0 \quad (4.3.6)$$

which, when solving for $p(n)$ and grouping terms for values of $\pm k$, gives the generalised pentagonal numbers. This gives Equation 4.3.2, proving Theorem 4.3.1.

□

Using this information, and the recurrence relation that can be derived from it, Srinivasa Ramanujan discovered the following congruences:

$$p(5k + 4) \equiv 0 \pmod{5} \quad p(7k + 5) \equiv 0 \pmod{7} \quad p(11k + 6) \equiv 0 \pmod{11}$$

These can be used to speed up the computation of large partitions, and are proved in (Ramanujan, S., 1921) which was extracted from the manuscripts of G. H. Hardy after death of Ramanujan in 1920.

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